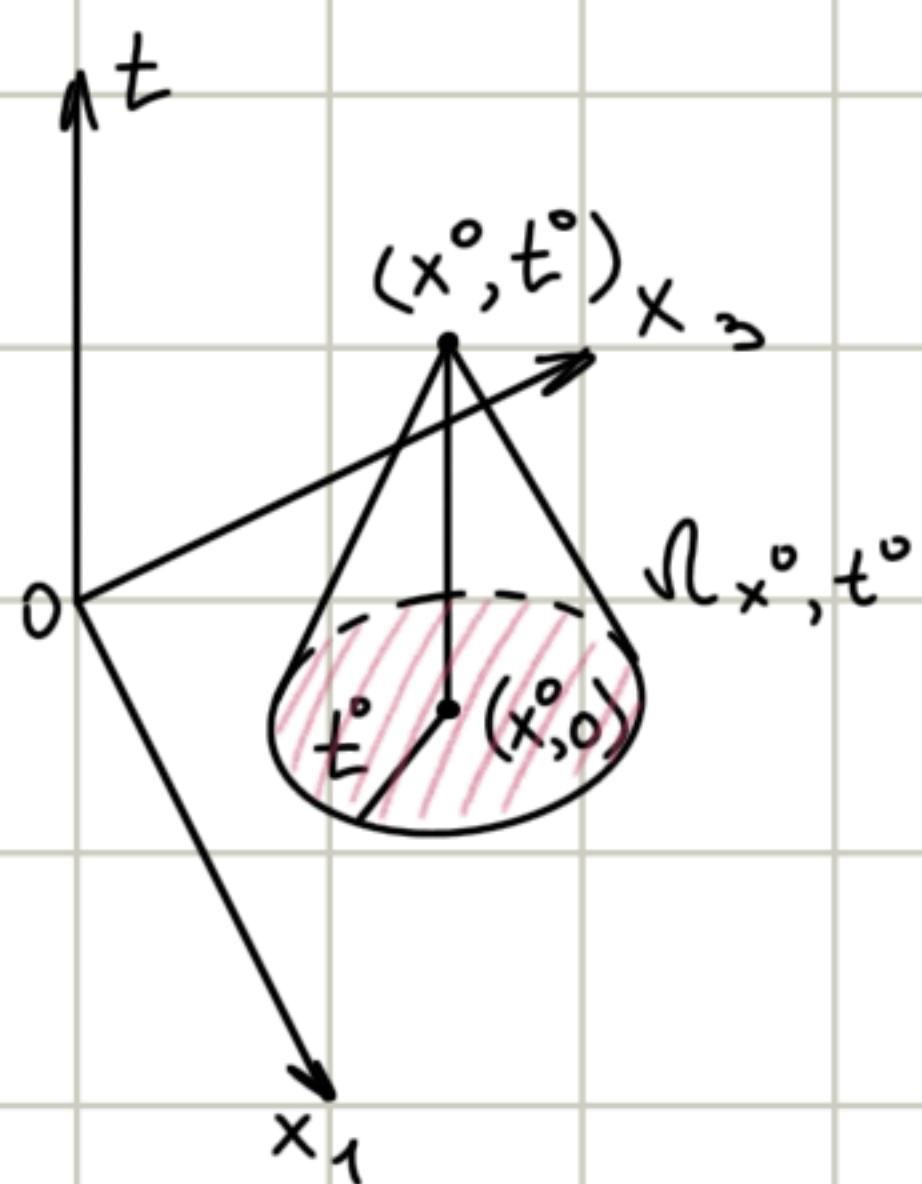




$$\bar{\Omega}_{x^0, t^0} = \Omega_{x^0, t^0} \cup \partial \Omega_{x^0, t^0}$$

бок. пов.
(есть кар-ка
вотм. ур.)

норм. кривизна = $\bar{\Omega}_{x^0, t^0}$

$$x = (x_1, x_2, x_3)$$

$$y = (y_1, y_2, y_3)$$

$$g(y), y \in \bar{\Omega}_{x^0, t^0} \rightarrow A_g(x, t) = \frac{1}{4\pi t} \iint_{|y-x|=t} g(y) dS_y = \frac{t}{4\pi} \iint_{|\eta|=1} g(\bar{x} + \bar{\eta}t) dS_{\eta}$$

1) условия

$$2) A_g(x, t)|_{t=0} = 0$$

$$3) \Delta_x (A_g(x, t)) = A_{\Delta g}(x, t)$$

$$4) \frac{\partial}{\partial t} (A_g(x, t))|_{t=0} = g$$

$$5) \frac{\partial^2}{\partial t^2} (A_g(x, t))|_{t=0} = 0$$

$$6) \frac{\partial^2}{\partial t^2} (A_g(x, t)) = A_{\Delta g}(x, t)$$

$$\begin{aligned} \frac{\partial}{\partial t} A_g(x, t) &= \frac{1}{4\pi} \iint_{|\eta|=1} g(\bar{x} + \bar{\eta}t) dS_{\eta} + \frac{t}{4\pi} \iint_{|\eta|=1} (\bar{\nabla} g, \bar{\eta}) dS_{\eta} = \frac{1}{t} \left(\frac{1}{4\pi t} \iint_{|y-x|=t} g(y) dS_y \right) + \\ &+ \frac{1}{4\pi t} \iint_{|y-x|=t} (\bar{\nabla} g, \bar{\eta}) dS_y = \frac{1}{t} A_g(x, t) + \frac{1}{4\pi t} \iint_{|y-x|<t} \Delta g(y) dy = \\ &= \frac{1}{t} A_g(x, t) + \frac{1}{4\pi t} \int_0^t dr \left(\iint_{|y-x|=r} \Delta g(y) dS_y \right) \end{aligned}$$

$$\begin{aligned} \frac{\partial^2}{\partial t^2} (A_g(x, t)) &= -\frac{1}{t^2} A_g(x, t) + \frac{1}{t} \left\{ \frac{1}{t} A_g(x, t) + \frac{1}{4\pi t} \int_0^t dr \left(\iint_{|y-x|=r} \Delta g(y) dS_y \right) \right\} - \frac{1}{4\pi t^2} \int_0^t dr \left(\iint_{|y-x|=r} \Delta g(y) dS_y \right) + \\ &+ \frac{1}{4\pi t} \iint_{|y-x|=t} \Delta g(y) dS_y \end{aligned}$$

$$\frac{\partial^2}{\partial t^2} (A_g(x, t)) = \frac{1}{4\pi t} \iint_{|y-x|=t} \Delta g(y) dS_y$$

$$A_{\Delta g}(x, t)$$

$$\frac{t}{4\pi} \iint_{|\eta|=1} \Delta g(\bar{x} + \bar{\eta}t) dS_{\eta} \xrightarrow{t=0} \frac{\partial^2}{\partial t^2} (A_g(x, t))|_{t=0} = 0$$

th

$$(1) u_{tt} - \Delta u = 0, x \in \mathbb{R}^3, t > 0$$

$$(2) u|_{t=0} = \varphi(x), x \in \mathbb{R}^3$$

$$(3) u_t|_{t=0} = \psi(x), x \in \mathbb{R}^3$$

$\varphi(x) \in C^3(\mathbb{R}^3), \psi(x) \in C^2(\mathbb{R}^2)$, тогда реш. задачи (1), (2), (3) $\exists u$

$$u(x, t) = A_{\varphi}(x, t) + \frac{\partial}{\partial t} (A_{\psi}(x, t))$$

нужно док-но

$$u(x, t) \in C^2 (x \in \mathbb{R}^3, t \geq 0)$$

усл. (1), усл. (2), усл. (3)

верно

н.к. * не нули *

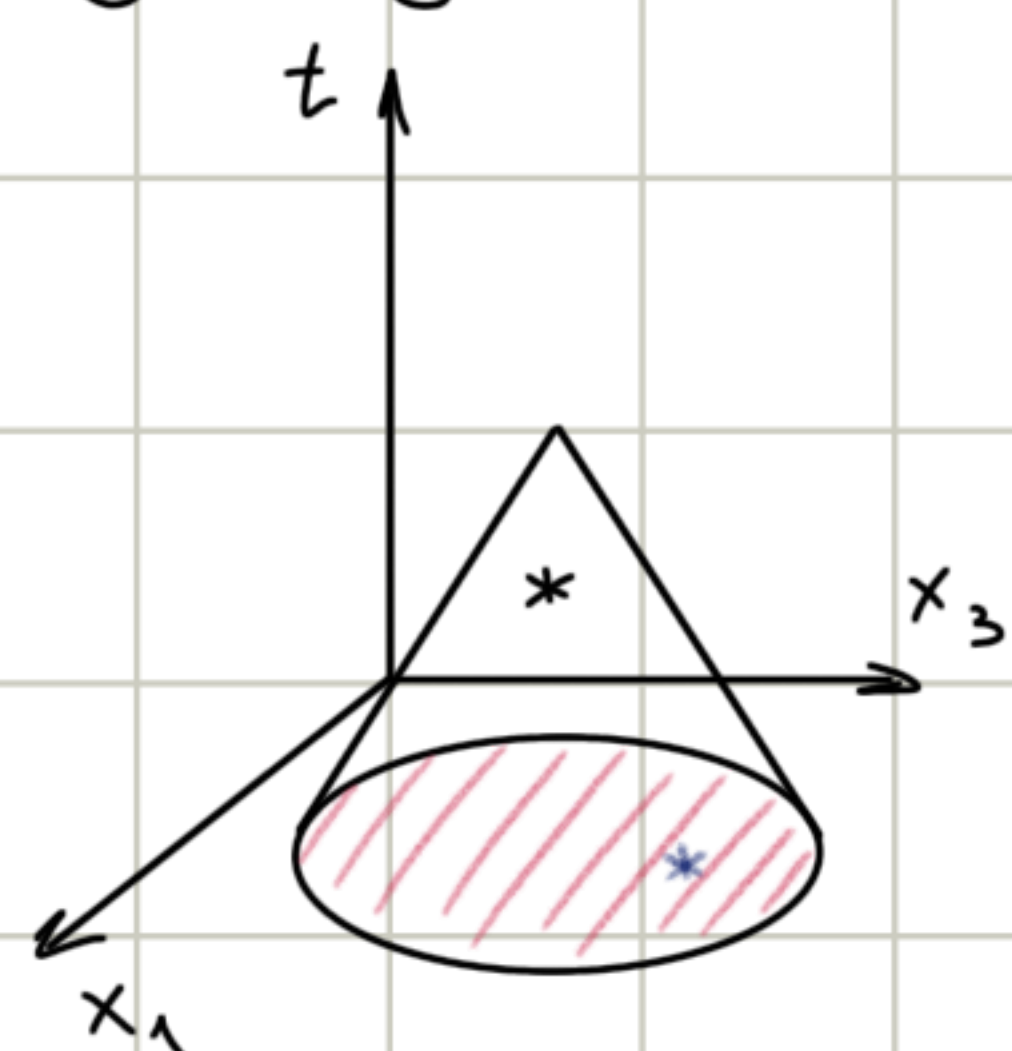
$$u(x, t) = A_{\varphi}(x, t) + \frac{\partial}{\partial t} (A_{\psi}(x, t))$$

$C^2(\bar{\Omega}_{x^0, t^0})$

но об-б-е

$C^3(\bar{\Omega})$

$$(2) u|_{t=0} = A_{\varphi}|_{t=0} + \frac{\partial}{\partial t} (A_{\psi})|_{t=0} = \varphi \Rightarrow \text{усл. (2)}$$



$$(3) u_t|_{t=0} = \frac{\partial}{\partial t} (A\psi)|_{t=0} + \frac{\partial^2}{\partial t^2} (A\psi)|_{t=0} = \psi \Rightarrow \text{yг. (3)}$$

$$u_{tt} = \underbrace{\frac{\partial^2}{\partial t^2} (A\psi)}_{A\Delta\psi = \Delta(A\psi)} + \underbrace{\frac{\partial^2}{\partial t^2} \left(\frac{\partial}{\partial t} (A\psi) \right)}_{\frac{\partial}{\partial t} \left[\frac{\partial^2}{\partial t^2} (A\psi) \right] = \frac{\partial}{\partial t} (\Delta(A\psi))} = \Delta(A\psi) + \frac{\partial}{\partial t} (\Delta(A\psi)) = \Delta(A\psi) + \Delta\left(\frac{\partial}{\partial t} (A\psi)\right) = \Delta\left(A\psi + \frac{\partial}{\partial t} (A\psi)\right) = \Delta(u) = \Delta(u)$$



$$u(x, t) = \frac{t}{4\pi} \iint_{|\eta|=1} \psi(\bar{x} + \bar{\eta}t) dS_{\eta} + \frac{1}{4\pi} \iint_{|\eta|=1} \varphi(\bar{x} + \bar{\eta}t) dS_{\eta} + \frac{t}{4\pi} \iint_{|\eta|=1} (\bar{\nabla} \varphi, \bar{\eta}) dS_{\eta} \quad (*)$$

вернемся позже

th

Задача

не может иметь более одного решения

$$\begin{cases} u_{tt} - \Delta u = F(x, t), & x \in \mathbb{R}^3, t > 0 \\ u|_{t=0} = \varphi(x), & x \in \mathbb{R}^3 \\ u_t|_{t=0} = \psi(x), & x \in \mathbb{R}^3 \end{cases}$$

где $\varphi(x), \psi(x), F(x, t)$ yг. необх. yсл.

Лемма

$$1) v(x, t) \in C^2(\bar{Q}_{x^0, t^0})$$

$$2) v_{tt} - \Delta v = 0 \text{ в } Q_{x^0, t^0}$$

$$3) v|_{t=0} = 0, x \in \bar{Q}_{x^0, t^0}$$

$$4) v_t|_{t=0} = 0, x \in \bar{Q}_{x^0, t^0}$$

$$1), 2), 3), 4) \Rightarrow v(x, t) \equiv 0 \text{ в } \bar{Q}_{x^0, t^0}$$

Δ. th.

$$u_1(x, t) \text{ и } u_2(x, t) - \text{реш.}$$

$$v(x, t) = u_1(x, t) - u_2(x, t)$$

$$v(x, t) \in C^2(x \in \mathbb{R}^3, t \geq 0)$$

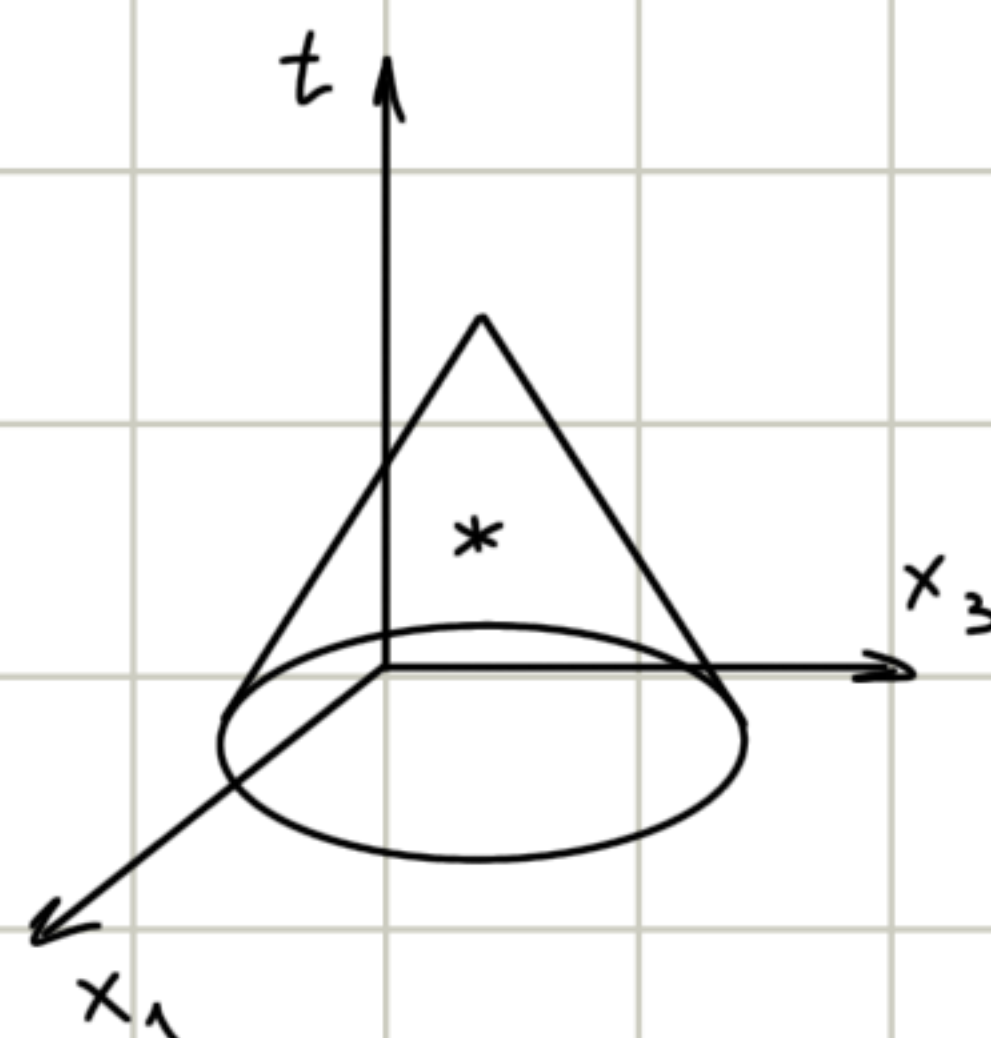
$$v_{tt} - \Delta v = 0, x \in \mathbb{R}^3, t > 0$$

$$v|_{t=0} = 0, x \in \mathbb{R}^3$$

$$v_t|_{t=0} = 0, x \in \mathbb{R}^3$$

$$\text{Цель: } v(x, t) = 0 \quad \forall x \in \mathbb{R}^3 \quad \forall t \geq 0$$

$$v(x, t) \begin{matrix} \swarrow t=0 \\ \searrow t>0 \end{matrix} \quad \begin{matrix} v(x, 0) = 0 \\ v_t(x, 0) = 0 \end{matrix}$$



Δ. Лемма

$$(\xi, \tau) \in \bar{Q}_{x^0, t^0}, \tau > 0$$

$$\text{Цель: } v(\xi, \tau) = 0$$

$$\bar{\mathcal{D}}_{\xi, \tau} \subset \bar{\mathcal{D}}_{x^0, t^0}$$

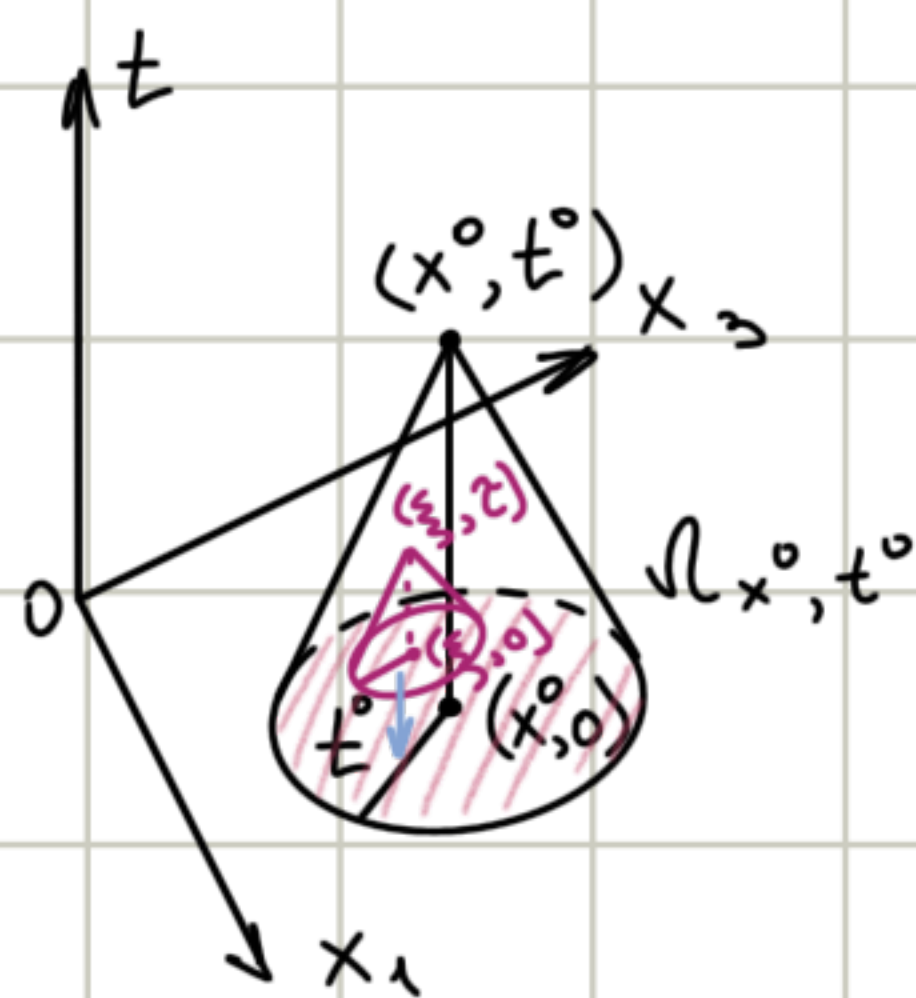
(маленький конус)

$$v(x, t) \in C^2(\bar{\mathcal{D}}_{\xi, \tau})$$

$$\text{в } \mathcal{D}_{\xi, \tau}: v_{ttt} - \Delta v = 0$$

$$v_{ttt}(x, t) - \Delta v(x, t) \equiv 0 \quad \forall (x, t) \in \mathcal{D}_{\xi, \tau}$$

$$\bar{A}(x, t), \operatorname{div} \bar{A} \equiv 0, \iiint_{\mathcal{D}_{\xi, \tau}} \operatorname{div} \bar{A} dx = 0 = \iint_{\partial \mathcal{D}_{\xi, \tau}} (\bar{A}, \bar{n}) dS$$



$$v_{ttt} - v_{x_1 x_1} - v_{x_2 x_2} - v_{x_3 x_3} \equiv 0 \quad | \cdot v_t$$

$$v_{ttt} v_t - v_{x_1 x_1} v_t - v_{x_2 x_2} v_t - v_{x_3 x_3} v_t \equiv 0$$

$$v_{ttt} \cdot v_t = \frac{1}{2} (v_t^2)'_t$$

$$(v_{x_1}^2)'_t = 2 v_{x_1} v_{x_1 t}$$

$$(v_{x_1} \cdot v_t)'_{x_1} = v_{x_1 x_1} \cdot v_t + v_{x_1} v_{x_1 t} = v_{x_1 x_1} v_t + \frac{1}{2} (v_{x_1}^2)'_t$$

$$v_{x_1 x_1} v_t = (v_{x_1} v_t)'_{x_1} - \frac{1}{2} (v_{x_1}^2)'_t$$

$$\frac{1}{2} (v_t^2)'_t - [(v_{x_1} v_t)'_{x_1} - \frac{1}{2} (v_{x_1}^2)'_t] - [(v_{x_2} v_t)'_{x_2} - \frac{1}{2} (v_{x_2}^2)'_t] - [(v_{x_3} v_t)'_{x_3} - \frac{1}{2} (v_{x_3}^2)'_t] \equiv 0$$

$$-\frac{\partial}{\partial x_1} (v_{x_1} v_t) - \frac{\partial}{\partial x_2} (v_{x_2} v_t) - \frac{\partial}{\partial x_3} (v_{x_3} v_t) + \frac{\partial}{\partial t} \left(\frac{v_{x_1}^2 + v_{x_2}^2 + v_{x_3}^2 + v_t^2}{2} \right) \equiv 0$$

$$t \rightarrow x_4$$

$$\bar{A}(x_1, x_2, x_3, x_4) = \begin{pmatrix} -v_{x_1} v_{x_4} \\ -v_{x_2} v_{x_4} \\ -v_{x_3} v_{x_4} \\ \frac{v_{x_1}^2 + v_{x_2}^2 + v_{x_3}^2 + v_{x_4}^2}{2} \end{pmatrix}$$

$$C^1(\bar{\mathcal{D}}_{x^0, t^0})$$

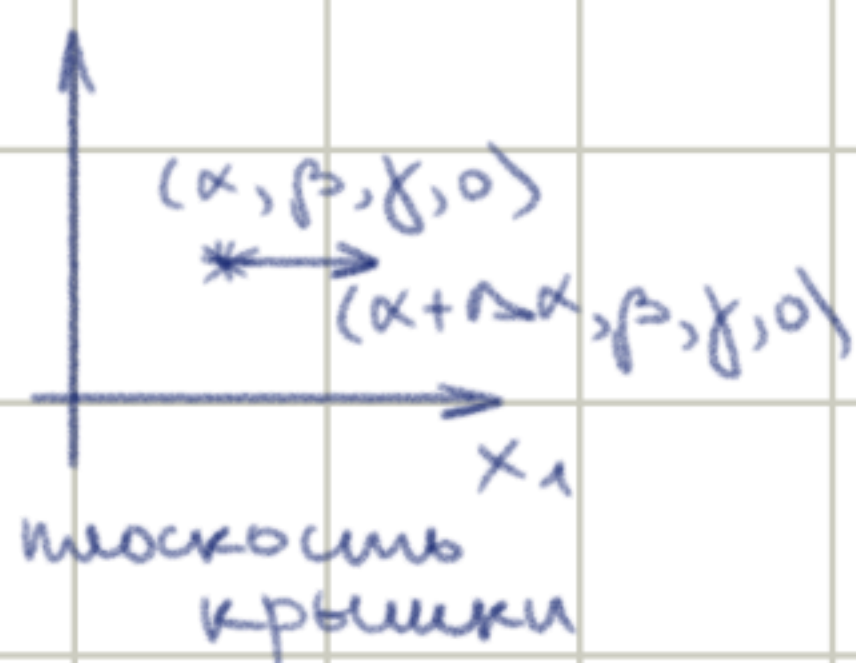
$$\operatorname{div} \bar{A}(x_1, x_2, x_3, x_4) \equiv 0 \quad \text{в } \mathcal{D}_{\xi, \tau}$$

$$0 = \iiint_{\mathcal{D}_{\xi, \tau}} \operatorname{div} \bar{A}(\bar{x}) d\bar{x} = \iint_{\partial \mathcal{D}_{\xi, \tau}} (\bar{A}(\bar{x}), \bar{n}) dS = \iint_{\text{Б.П.}} (\bar{A}, \bar{n}) dS + \iint_{\text{Н.кр.}} (\bar{A}, \bar{n}) dS = \iint_{\text{Б.П.}} (\bar{A}, \bar{n}) dS$$

$$\bar{n} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

на нижн. крышке

$$(\bar{A}, \bar{n}) = -\frac{1}{2} (v_{x_1}^2 + v_{x_2}^2 + v_{x_3}^2 + v_{x_4}^2) = 0$$



$$v'_{x_1}(\alpha, \beta, \gamma, 0) = \lim_{\Delta \alpha \rightarrow 0} \frac{v(\alpha + \Delta \alpha, \beta, \gamma, 0) - v(\alpha, \beta, \gamma, 0)}{\Delta \alpha}$$